International Review of Financial Analysis 99 (2025) 103932



Contents lists available at ScienceDirect

International Review of Financial Analysis

journal homepage: www.elsevier.com/locate/irfa





Can tax reduction incentive policy promote corporate digital and intelligent transformation?

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ARTICLE INFO

Keywords:
Tax reduction incentive
Digital transformation
Resource complementarity
Expected cash flow

ABSTRACT

Digital intelligence has provided companies a great opportunity to develop a distinct and differentiating competitive advantage. Using the difference-in-differences approach, this study investigated the relationship between implementing tax reduction incentives policies and the digitalization and intelligence of A-share listed companies from 2016 to 2022. Results show that implementing a tax reduction incentive strategy can significantly improve the degree of digitalization and intelligence that companies display. Furthermore, financial flow systems and resource complementarity are expected to contribute to positive results. By encouraging human resource development and providing financial support through resource expansion, tax reduction incentives have assisted businesses in digital transformation. Concurrently, the tax reduction incentive policy helps enterprises in stabilizing their expected cash flow, reducing their financing challenges, and providing critical assurance for the acceptance and development of digital technology. This helps the company transition to digital intelligence. Moreover, the degree of urban digital infrastructure, the type of industry, and companies' reliance on external financing exhibit different mediating effects. The study results have practical relevance for companies' digital and intelligent transformation.

1. Introduction

Global digital economic development is accelerating, which is critical for restructuring global factor resources, redesigning the global economic structure, and changing the global competition landscape. Enterprise digital intelligence transformation entails optimizing production operations and monitoring various aspects of procurement, production, inventory, capital, quality, energy consumption, and equipment status, allowing enterprises to address complex problems and improve product quality for new product innovation (Zhang, Bai, Wang, & He, 2024). However, transforming enterprise digital intelligence requires a multifaceted investment strategy that includes upgrading technology and equipment, establishing digital management systems, restructuring human resources, and transforming business models. This process incurs significant investment costs. Corporate funding constraints are imposed by law and by the need to maintain employment. In reality, a problem exists with businesses being resistant to change, lacking the courage to change, and struggling to implement successful transformations.

Thus, following the theory of market failure, government action is required to close the investment gap in firms' digital and intelligent transformation. As the visible hand, the government is currently catalyzing the transformation of micro-entities into firms through digital and intelligent transformation. It may provide a comprehensive policy toolkit to assist firms with digital and intelligent transfer. Tax incentives have broad applicability and upfront incentives, as well as well-defined policy objectives and high policy effectiveness, thus improving their ability to support firms in their digital and intelligent transformation. Tax incentives can take many forms, including exemptions and reductions for technology transfers, preferential tax rates for technologically advanced enterprises, additional deductions for research and development expenses, and accelerated depreciation of fixed assets due to technological transformation and equipment renewal (Wang & Qin, 2024). These incentives are consistent with digital transformation concepts, such as digital investment, technological, and business model transformations. Furthermore, these incentives are extremely compatible with each other. However, tax incentives can alleviate the financial burden on enterprises, assist in their transformation, and influence their behavior by fostering innovation and entrepreneurship. These factors can jointly assist enterprises in transitioning to digital intelligence.

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https://doi.org/10.1016/j.irfa.2025.103932
Received 2 September 2024; Received in revised form 13 January 2025; Accepted 14 January 2025
Available online 17 January 2025
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There is a lack of literature that explores the relationship between tax cuts and intelligent transformation. Dai and Chapman (2022) utilize a theoretical model to examine the impact of tax and fee reduction policies on the intelligent upgrading of manufacturing in the context of flexible production. The study finds that these policies can promote intelligent upgrading in the short term. However, it suggests that excessively rapid tax and fee reductions may limit their effectiveness. Note that he study does not investigate the effects of specific tax reduction policies or provide empirical evidence. According to additional research, the primary factors influencing the intelligent upgrading of firms are the development of intelligent technology (Feng, Wang, Song, & Liu, 2022; Xu, Guo, Yue, & Li, 2024), technological innovation (Guan, Wang, & He, 2024), labor costs (Huang & Liu, 2024), environmental regulations, policy incentives, product market demand, and entrepreneurial, innovative spirit (Jia, Ding, & Liu, 2020). However, studies on tax policy and corporate behavior show that implementing tax reduction policies can encourage businesses to invest in fixed assets (Yang, Deng, & Du, 2023), reorganize, and improve the quality of their human resources. Many scholars have utilized the opportunities provided by value-added tax (VAT) reform and income tax reduction policies to conduct empirical studies on the effect of tax reduction reforms (Zhou, Shi, Bao, Gao, & Ma, 2023). These reforms include VAT reform, camping reform, tax credit refund, fixed asset accelerated depreciation, and tax incentives (Lu, Liu, Cao, & Li, 2023). Using various levels of data and methodologies, the studies examined the impact of these reforms on investment, innovation, and human capital (Qi, Zhang, & Chen, 2023). Implementing tax reduction measures, such as switching to a VAT system and reforming camps, can boost fixed asset investment and innovation (Hao, Peng, & He, 2023). The camp reform enhances enterprises' fixed asset investment and R&D investment by substituting non-debt tax shelters for capital structure (Sun, Zhan, & Du, 2020). The U.S.'s fixed asset depreciation tax incentives encourage investment in fixed assets, with small enterprise showing a more pronounced response than large firms. China's accelerated depreciation policy primarily boosts investment in small-scale businesses and privately owned organizations, while also significantly affecting capital allocation efficiency within these firms. Implementing VAT transition and camp reform improves the organic composition of enterprises' capital and facilitates human capital development (Ullah, Niu, & Meo, 2024). The tax credit rebate program encourages businesses to increase their investment in fixed assets by removing financial constraints. Enterprises see more outstanding VAT credit refunds, increasing their investment and, ultimately, contributing to the growth of their overall value.

Hence, little literature offers evidence for the positive relationship between tax reduction policies and firm intelligence transformation. There are two primary causes for this phenomenon: first, the government must continue implementing preferential policies aimed at facilitating firms' digital intelligent transformation. However, the tax cut strategy has undeniably impacted firms' digital and intelligent transformation. In today's digital economy, intelligent investment is critical to firm investment. It necessitates an optimal adjustment of capital and labor allocation. The tax cut reform encourages fixed asset and R&D investment, as well as the development of human capital and improving capital organic composition, thereby promoting firms' transition to digital intelligence (Zheng & Zhang, 2021). Second, the concept of enterprise smart manufacturing is based on the unique features of manufacturing technology, which makes it difficult to understand the fundamental aspect of the tax reduction policy's impact on enterprise digital intelligent transformation (Zhou, Zhou, & Ji, 2022). This paper examines the impact of tax incentives on the transformation of digital intelligence. The study contributes to academic innovation by establishing a theoretical foundation for assessing the effect of tax reduction policies on digital intelligent transformation. This research is extremely important because it aids in the design of tax reduction policies and promotes digital intelligent transformation and upgrading.

This study contributes to the current literature in two ways. The majority of research has focused on the economic implications of enterprise digital transformation in the context of the digital economy. Increased emphasis on the factors that drive corporate digital transformation is required. This study analyzes the fundamental factors influencing the digital transformation of businesses, providing insights into the existing literature on the subject. Second, although much literature explores the economic impacts of tax reform policies at micro and macro levels, further research is required on the effects of policy support, precisely tax cuts, on firms' digital transformation. This study demonstrates that tax reduction incentives effectively promote the transformation of a company's numerical intelligence by supplying essential human capital and financial resources for this process. This analysis evaluates the effectiveness of these incentives through the lenses of resource complementarity and anticipated cash flow.

2. Theoretical hypothesis

2.1. Direct effect of tax reduction incentives policy on firms' digitization and intelligence

Frictions in the capital market raise the cost of obtaining finance from external sources, limit enterprise funding, and prevent enterprises from meeting their investment targets. Furthermore, the digital transformation of enterprises involves activities that require a significant amount of capital (Zhou et al., 2022). The success or failure of these activities is highly uncertain, and the projects are often backed by low-value collateral, making it difficult for enterprises to secure sufficient funds through external financing (Guo, Li, Qiao, & Qi, 2023). The lack of funding impedes the progress of the digital transformation process. Tax breaks will help businesses transform into digital enterprises by mitigating their financial barriers. Moreover, taxes, as mandatory expenses for enterprises, result in resource outflow and decrease available internal funds. Conversely, tax incentives augment the enterprises' funds without imposing additional conditions on their business decisions, providing increased motivation for enterprises to engage in digital intelligent transformation (Zheng & Zhang, 2021). Second, under the principle of preferential financing, firms should prioritize internal financing due to its lower costs. Exogenous financing should only be considered if internal funding is insufficient (Qi et al., 2023). However, if the tax rate is excessively high, businesses will increasingly rely on external financing methods such as borrowing or issuing stocks, resulting in higher funding costs. Furthermore, they will have to bear the financial burden of repaying the principal amount upon maturity and making regular interest or dividend payments. Such conditions are not conducive to implementing transformative activities. Tax advantages enable businesses to obtain reliable internal finances, resulting in lower financing costs (Lu, Wang, & Wang, 2023). Moreover, tax incentives effectively reduce enterprises' financial constraints, allowing them to allocate more resources to transformative areas and contribute to their digital transformation, as evidenced by improved cash flow and lower financing costs.

Furthermore, tax incentives can encourage enterprises to upgrade their equipment. As a critical component of firm investment, equipment upgrading demonstrates enterprises' willingness to explore new domains and actively pursue transformation. Tax reduction incentives play a crucial role in increasing company investment expenditures by lowering the cost of capital for users (Feng et al., 2022). Furthermore, they result in changes in investment opportunities. Consequently, the tax incentive level directly correlates with investment willingness. Under the current policy framework, tax incentives, such as investment credits and accelerated depreciation, can assist enterprises in improving resource allocation efficiency, reducing investment and renewal costs, increasing equipment replacement frequency, and facilitating technological advancement. Moreover, these incentives alleviate financial constraints (Liang, Liang, & Tang, 2024). Enhancing a company's digital intelligence necessitates allocating increased resources to innovation

(Ullah et al., 2024). However, innovation activities are inherently risky and uncertain regarding potential returns, reducing enterprises' motivation to invest in innovation (Tian, Li, & Cheng, 2022). Tax advantages to firms by the government can offset the costs of innovation and contribute to the expected increase in profits resulting from digital transformation. This encourages enterprise to augment their investment in innovation. Due to tax incentive legislation, firms are increasingly motivated to embark on digital transformation as it leads to increasing business outcomes. The tax incentive encourages organizations to engage in technology innovation, pursue transformation, and take risks, thereby facilitating businesses' digital and intelligent transformation. Thus, this study proposed the following hypothesis.

Hypothesis 1: Tax reduction incentives policy can effectively improve firms' digitization and intelligence level.

2.2. Mechanisms of tax reduction incentives policy on firms' digitization and intelligence

According to the concept of the resource complementary effect, adding high-quality labor to production equipment can result in a more substantial complementary effect, increasing firm demand for high-skilled labor and boosting enterprise human capital. Enterprise digital transformation is a process that involves internal technology change. The skills, knowledge, and abilities of employees significantly impact the effectiveness of this transformation (Sun et al., 2020). Upgrading human capital will significantly impact the success of enterprise digital transformation. The first approach is skill matching, which involves increasing employees' skill levels. Organizations can ensure that their employees can adapt to digital tools and technologies, allowing them to effectively support the digital intelligence transformation (Dai & Chapman, 2022). It also addresses changes in internal management. Digital transformation frequently necessitates interdepartmental collaboration and information exchange, resulting in organizational changes (Guo, Wei, Wang, & Hu, 2024). Enhancing human capital entails increasing employees' ability to adapt to change, encouraging collaboration with other departments, and providing them with the necessary skills and abilities to thrive in the digital era, thereby facilitating improved digital transformation. Thus, this paper proposed the following hypothesis:

Hypothesis 2: Tax reduction incentives policy can effectively improve firms' levels of digitization and intelligence via resource complementarity mechanisms.

The tax reduction policy has effectively reduced the burden of corporate taxes, increased the amount of cash that firms can retain, and improved internal financial support for the transition to digital intelligence (Ye & Yue, 2024). Digital intelligence transformation is a complex and ongoing process that aims to increase efficiency and optimize management within a company. Moreover, its success or failure is uncertain and will only be determined with time (Sun et al., 2020; Yue, Korkmaz, Yin, & Zhou, 2022). Digital transformation necessitates a significant financial investment from enterprises, as well as high profitability and consistent cash flow expectations, which are critical for protecting the capital invested in the transformation process and ensuring the enterprise's successful implementation of digital transformation (Lu, Wang, & Wang, 2023). Tax reduction incentives can reduce the enterprise's tax burden, giving it more money (Ran, Wang, Sun, & Qu, 2024). Furthermore, these incentives can boost corporate productivity, increase profitability, improve future cash flow expectations, and ensure capital investment in the enterprise's digital transformation (Ullah et al., 2024). Simultaneously, sophisticated machinery, procedures, and technologies can lead to highly productive manufacturing, while a skilled workforce can ensure superior production quality (Wang, Wan, Ma, Zhou, & Chen, 2023). This effectively lowers the additional cost of production, increases profitability, stabilizes cash flow expectations, and provides a tangible foundation for enterprises to transition to digital intelligence. Thus, this study proposed the following hypothesis.

Hypothesis 3: Tax reduction incentives policy can effectively improve firms' level of digitization and intelligence through cash flow mechanisms.

3. Research design

3.1. Variable selection

3.1.1. Measuring tax reduction incentives

On May 1, 2016, the pilot scope of the Business Continuity Management policy was expanded to include the construction, real estate, finance, and living services sectors. The expansion includes eliminating the business tax, extending the VAT, and fully implementing the VAT deduction process. VAT deductions now include real estate, intangible assets, and outsourced services. This expansion has reduced the costs of firms' intelligent investment projects and increased the availability of input tax credits.

The Ministry of Finance and the State Administration of Taxation have issued reform plans in response to the comprehensive camping reform, focusing on simplifying and lowering VAT rates. The VAT rate for manufacturing and other businesses fell from 17 % to 16 % on May 1, 2018. Meanwhile, transportation, construction, essential communications services, other sectors, and agricultural products saw their VAT rates reduced from 11 % to 10 %, demonstrating the reform's broad impact.

The reduction in VAT rates impacts organizations' strategic transformations in terms of resource utilization and outcome attainment. It facilitates cost reduction and reduces cash outflows associated with strategic investments by organizations, resulting in a shorter payback period, and improves the viability of investment decisions for intelligent transformation. Following two consecutive VAT rate reductions, the input tax on investments in intelligent equipment and customized software can be reduced by 4 % after 2019. The input tax on real estate investments required for new intelligent workshops or digital factories can be reduced by 2 %, decreasing enterprise investment overall. It also promotes the transition to a service-oriented model in the manufacturing sector and lowers the overall output tax burden. Adjusting the VAT rate reduces the tax rate that businesses must pay when selling goods, real estate, intangible assets, and services. Furthermore, enterprise informatization and digitization have improved significantly due to enterprises' intelligent transformation and upgrading. This enhancement supports the manufacturing sector's service-oriented and informatization transformation, resulting in restructuring enterprise income. Implementing a tax rate of 13 % on goods sales and 6 % on service income can significantly reduce enterprises' tax burden. Reducing the output tax diminishes the VAT burden on businesses, resulting in tax savings.

Therefore, this study introduces an interaction, represented as $tax_{it} = treat_i \times post_t$.to accurately assess the impact of tax reduction incentives policy. The dummy variable $post_t$. $post_t$ is assigned a value of 1 in and after 2018, and 0 otherwise. Adjusted industry firms are referred to as the treatment group ($treat_i = 1$), whereas the other group is referred to as the control group ($treat_i = 0$).

3.1.2. Measuring firms' level of digitization and intelligence

The enterprise's annual report database was used to create a thesaurus of 29 keywords related to the enterprise's digital and intelligent transformation, such as intelligent, intelligent manufacturing. Python was used to extract content for text analysis as well as negative keyword processing. To determine an enterprise's digital and intelligent development level (*digital*), this paper compares the proportion of annual keywords used by the enterprise to the proportion of comparable keywords used by other enterprises in the same industry during the same year.

3.1.3. Control variables

This study selected three control variables from three dimensions: financial, governance, and characteristics. Given the apparent relationship between firms' financial condition and income, which impacts labor income allocation, this study selects the following indicators to measure the financial dimension: leverage (*LEV*), return on assets (*ROA*), and development ability (*GROWTH*). An analysis of the influence of corporate governance elements on management motivation and, therefore, the allocation of labor revenue: The independent director (*IND*) and largest shareholder (*TOP1*) ratios. Furthermore, significant disparities in worker remuneration exist among companies that go through multiple life cycles. Thus, this paper includes firm size (*SIZE*) and firm age (*AGE*) as control variables. Table 1 shows the detailed definitions.

3.2. Methodology

This study uses the difference-in-differences (DID) model to investigate the impacts of tax reduction incentives policy on corporate digitization and intelligence, as follows:

$$digital_{it} = \alpha + \beta \cdot tax_{it} + \sum \gamma \cdot X + year_t + firm_i + \varepsilon_{it} \quad (1)$$

$$M_{it} = \alpha + \beta \cdot tax_{it} + \sum \gamma \cdot X + year_t + firm_i + \varepsilon_{it} \quad (2)$$

3.3. Data sources

Given that the entire campground change will take place in March 2016, this paper chooses a sample period spanning 2016 to 2022 to accurately analyze the policy effects. The sample included 22,121 observations of Chinese A-share-listed firms. The data used in this study came from the China Stock Market & Accounting Research Database. Similarly, the descriptive statistics of the variables, as shown in Table 1, show that all variables fell within the expected range of values.

Table 1
Descriptive statistics.

Variables	Definition	Obs	Mean	Std.	Min	Max
digital	Firms' level of digitization and intelligence	21,805	2.693	1.292	0.000	7.266
tax	Tax reduction incentives	22,121	0.472	0.499	0.000	1.000
SIZE	The natural logarithm of a company's total assets	22,121	22.383	1.327	19.987	26.408
AGE	The age of the company	22,016	2.262	0.826	0.000	3.497
ROA	Return on assets	22,121	0.032	0.073	−0.335	0.204
LEV	The proportion of overall loans to total assets	22,121	0.428	0.199	0.067	0.909
FIX	the rate of revenue growth	22,121	0.199	0.152	0.002	0.668
GROWTH	Fixed assets ratio	22,017	0.337	0.853	−0.765	5.832
SOE	Enterprises' ownership	22,121	0.326	0.469	0.000	1.000
IND	The ratio of independent directors	22,121	0.379	0.056	0.000	0.800
TOP1	The ratio of largest shareholder	22,121	0.329	0.147	0.018	0.900

4. Empirical analysis

4.1. Baseline regression

Table 2 shows the estimation regression results. The estimated coefficients have positive effects at the 1 % significance level, indicating that implementing a tax reduction incentives policy can improve firms' levels of digitization and intelligence, regardless of whether the control variable was included in the analysis.

Columns (2)–(4) show the results of regressions that include the appropriate control variables. Notably, the estimated coefficients for firms' revenue growth rate (*GROWTH*) have no relationship with firms' level of digitization and intelligence, implying that firms' revenue growth rate is not a critical factor in their level of digitization and intelligence. Furthermore, the coefficient for the largest shareholding (*TOP1*) is significantly negative at 1 %, suggesting that a higher level of equity concentration impedes corporate digitization and intelligence. The finding focuses on the necessity and importance of equity diversification.

4.2. Robustness check

4.2.1. Parallel trend tests

The DID approach is based on the assumption that if the treatment group did not receive any policy intervention, its trend would be the same as the control group. This assumption is known as the parallel trend assumption. Therefore, this paper adopts dynamic events analysis as follows:

$$\begin{aligned} digital_{it} = {} & \alpha + \beta_{-2} * treat_i * post_t^{-2} + \beta_{-1} * treat_i * post_t^{-1} + \beta_0 * treat_i * post_t^{0} \\ & + \beta_{+1} * treat_i * post_t^{+1} + \beta_{+2} * treat_i * post_t^{+2} + \sum \gamma \cdot X + year_t \\ & + firm_i + \varepsilon_{it} \end{aligned} \quad (3)$$

The variable $post_t^j$ is a binary indicator representing the period relative to the year when the tax reduction incentives policy was

Table 2
Baseline regression results.

	(1)	(2)	(3)	(4)
	digital	*digital*	*digital*	*digital*
tax	0.314***	0.272***	0.315***	0.283***
	(0.045)	(0.044)	(0.043)	(0.043)
SIZE		0.065***	0.103***	0.133***
		(0.016)	(0.018)	(0.018)
AGE		−0.305***	−0.301***	−0.299***
		(0.027)	(0.028)	(0.029)
ROA			−0.361*	−0.397**
			(0.203)	(0.199)
LEV			−0.307***	−0.303***
			(0.116)	(0.115)
FIX			−1.281***	−1.124***
			(0.120)	(0.121)
GROWTH			−0.025	−0.020
			(0.019)	(0.018)
SOE				−0.124**
				(0.049)
IND				0.609
				(0.288)
TOP1				−0.896***
				(0.137)
cons	2.544***	1.808***	1.332***	0.739*
	(0.030)	(0.352)	(0.367)	(0.380)
controls	NO	YES	YES	YES
FE	YES	YES	YES	YES
N	21,804	21,700	21,636	21,636
R^2	0.172	0.196	0.215	0.227

Note: The robustness standard errors of the coefficients are in parentheses; ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

implemented in 2018. When the value of j is negative, it indicates the number of years prior to the implementation of the tax reduction incentives policy. When the value of j is greater than zero, it represents the year following the implementation of the tax reduction incentive policy. The reference year for this study is 2017, which occurred before the tax reduction incentives policy was implemented. The coefficient β measures the difference in corporate digitization and intelligence between treatment and control groups in year j, compared to the difference in the base year.

Table 3 presents the coefficient estimates for β_j and their confidence intervals at a 90 % level. Table 3 shows that during the initial years of implementing tax reduction incentives policy, the estimated coefficients of β_{-2} through β_0 do not pass the 10 % level of significance. Between 2019 and 2020, the estimated values of β_1 to β_{+2} are all statistically positive at the 10 % level. The findings show that there was no significant difference in corporate digitization and intelligence between the treatment and control groups prior to the implementation of the tax reduction incentives policy, which meets the parallel trend tests. Compared to the estimated value β_{+1}, the value of β_{+2} presents more significant, indicating that the tax reduction incentives policy has ongoing effects on promoting corporate digitization and intelligence.

4.2.2. Propensity score matching DID

This paper uses the propensity score matching technique to pair control group cities with similar features to treatment group cities to address the impact of observable variable variations and mode errors. First, the specific method involves using the control variables from the baseline regression as the characteristic variables, such as firm size (*SIZE*), age (*AGE*), growth ability (*GROWTH*), corporate governance (*TOP1* and *IND*), ownership (*SOE*), asset structure (*LEV*), return on assets (*ROA*), and fixed assets ratio (*FIX*), and calculating the propensity score values using the Logit model. The nearest neighbor matching technique then selects a firm as the control group for each treatment group based on the propensity score value. Finally, the corresponding samples underwent multiple regression analysis. Column (1) of Table 4 contains the result. The estimated coefficient is significantly positive at the 1 % level, which supports the benchmark regression results.

4.2.3. Placebo tests

The benchmark results show that the tax reduction incentive policy significantly and positively affects firms' digitization and intelligence. However, other macroeconomic policies or unpredictable circumstances may influence this effect. Thus, this study conducts a placebo test.

This study begins by randomly selecting an equal number of enterprises, using the treatment group sample distribution as a selection basis, to confirm the improvement in firms' digitization and intelligence owing to the implementation of the tax reduction incentives policy. Subsequently, a DID approach uses eq. (1) to generate coefficient estimates. The sampling procedure described above is repeated 1000 times, yielding 1000 estimated coefficients for DID analysis. The estimated coefficients of DID have a mean value of 0, which differs significantly from the coefficients of the benchmark regression. The findings indicate that other macroeconomic policies or random causes do not influence the study's estimated results. The improvement in firm digitization and intelligence is a result of the impact of implementing the tax reduction incentives policy.

Table 3
Parallel trend tests.

	digital
$treat_i \times post_t^{-2}$	0.034
	(0.026)
$treat_i{*}post_t^{-1}$	0.029
	(0.030)
$treat_i{*}post_t^{0}$	0.022
	(0.034)
$treat_i{*}post_t^{+1}$	0.069*
	(0.037)
$treat_i{*}post_t^{+2}$	0.103***
	(0.038)
cons	2.876***
	(0.605)
controls	YES
FE	YES
N	21,631
R^2	0.856

Note: The robustness standard errors of the coefficients are in parentheses; ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

4.2.4. Other robustness tests

First, alternative metrics for corporate digitization and intelligence should be adopted. This paper uses textual analysis to estimate the artificial intelligence (AI) level. Column (2) in Table 4 displays the result, which confirms the baseline regression.

Second, consider using an alternative measurement for tax breaks. Tax burdens (incentives) are utilized as a measure of tax reduction incentives, and they can be calculated by dividing the total tax rebates received by the firm by the sum of all tax refunds received and taxes paid by the enterprise. The magnitude of the tax incentive is directly proportional to its strength. Column (3) of Table 4 contains the results. The estimated coefficients are positive at the 1 % level, which confirms the benchmark regression results.

Third, consider specific sectors. Due to unique industry characteristics, intangible assets associated with company digital transformation are common in specialized industries such as the Internet, software, information technology, and other high-tech service sectors. Consequently, this study excludes four sectors, namely, Internet and related services, software and information technology services, professional and technical services, and science and technology promotion and application services, to reduce the influence of individual businesses in the benchmark regression. Column (4) of Table 4 presents the findings. The computed coefficient is significantly positive at 1 %, suggesting that tax reduction incentives continue to positively impact enterprise digitization and intelligence even when businesses characterized by digital transformation are excluded.

Fourth, considering the disruptions owing to the COVID-19 pandemic to organizational operations, this study omits the period from 2020 to 2022 to mitigate the impact of the COVID-19 pandemic. Column (5) of Table 5 presents the results and validates the benchmark findings.

Fifth, management decisions lag due to an overly complicated decision-making process, low-developed information technology, and insufficient human resources. To address the issue, we implemented a one-period lagged tax incentive policy. Column (6) in Table 4 reports the results, confirming the baseline regression results.

Finally, this study uses various fixed-effect controls. Column (7) of Table 4 shows the results consistent with the baseline regression results.

5. Further analysis

5.1. Mechanism test

5.1.1. Mechanisms for resource complementarity

To implement digital transformation, firms must incorporate advanced technological production equipment, necessitating aligning human and financial resources to achieve synergistic results. Acquiring advanced production equipment necessitates financial backing, which increases the demand for a skilled and specialized workforce. A company's strategy to accelerate its digital intelligence transformation must include financial assistance and the improvement of human resources. Therefore, this article uses the ratio of the total amount of cash paid for the construction of fixed assets, intangible assets, and other long-term

Table 4
Robustness tests results.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	digital	*AI*	*digital*	*digital*	*digital*	*digital*	*digital*
tax	0.283***	0.151***		0.655***	0.311***		0.283***
	(0.043)	(0.033)		(0.043)	(0.045)		(0.022)
L.tax						0.270***	
						(0.043)	
incentive			0.663***				
			(0.083)				
cons	0.836**	0.285***	0.918**	0.299***	0.504***	0.983***	0.739***
	(0.374)	(0.026)	(0.380)	(0.035)	(0.042)	(0.389)	(0.175)
controls	YES	YES	YES	YES	YES	YES	YES
FE	YES	YES	YES	YES	YES	YES	YES
Firm-year FE	NO	NO	NO	NO	NO	NO	YES
N	21,596	15,530	21,636	20,119	12,152	18,316	21,636
R^2	0.226	0.180	0.230	0.246	0.212	0.218	0.227

Note: The robustness standard errors of the coefficients are in parentheses. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 5
Mechanism test results: Resource complementarity.

	(1)	(2)	(3)	(4)
	rd	*digital*	*capital*	*digital*
tax	0.662***	0.136***	0.006***	0.281***
	(0.039)	(0.023)	(0.001)	(0.022)
rd		0.027***		
		(0.001)		
capital				0.344*
				(0.200)
cons	25.830***	0.111***	−0.019***	0.749***
	(3.569)	(0.002)	(0.006)	(0.175)
controls	YES	YES	YES	YES
FE	YES	YES	YES	YES
N	19,155	18,923	21,951	21,636
R^2	0.434	0.289	0.241	0.227

Note: The robustness standard errors of the coefficients are in parentheses. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 6
Mechanism test results: Expected cash flow.

	(1)	(2)	(3)	(4)
	cash	*digital*	*SA*	*digital*
tax	0.021***	0.283***	−0.012***	0.282***
	(0.004)	(0.022)	(0.001)	(0.022)
cash		0.232***		
		(0.075)		
SA				−0.076**
				(0.038)
cons	0.399***	0.642***	5.224***	1.137***
	(0.018)	(0.177)	(0.049)	(0.274)
controls	YES	YES	YES	YES
FE	YES	YES	YES	YES
N	21,951	21,636	21,950	21,636
R^2	0.297	0.227	0.372	0.227

Note: The robustness standard errors of the coefficients are in parentheses. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

assets to the enterprise's total assets to measure the enterprise's financial support (*capital*). The human capital structure of a business is measured by the ratio of R&D personnel to the total number of employees (*rd*).

Table 5 reports the results. The estimated coefficients are significantly positive at the 10 % level, indicating that tax-cutting policies can promote human resource structure, technology bias, and capital support, resulting in corporate digitization and intelligence.

5.1.2. Mechanism of expected cash flow

The digital intelligence transformation, which involves technological innovation within organizations to improve efficiency and optimize management, is a long-term battle. It is especially important for enterprises with strong profitability and stability as a foundation. As a result, this article uses the ratio of the company's monetary funds and trade financial assets to its total assets to determine its current cash flow (*cash*). Furthermore, the SA index evaluates the company's financial constraints (*SA*).

Table 6 presents the results. The estimated coefficients in column (1) of Table 6 are significantly positive at the 1 % level, implying that tax-cutting incentives can reduce corporate tax burdens, improving firms' cash flow. A sufficient cash flow can promote corporate digitization and intelligence through capital support, as shown in column (2) of Table 6. Furthermore, the addressed coefficients in column (3) of Table 6 are significantly negative at the 1 % level, indicating that tax-cutting policies can effectively alleviate corporate financing constraints, promoting corporate digital and intelligent transformation.

5.2. Heterogeneous analysis

5.2.1. The mediation of external financing dependence

The external environment has a significant impact on the production and operations of organizations. A favorable external environment can reduce enterprises' innovation time, decrease institutional transaction expenses, and improve enterprise resource allocation. Financial constraints may impede an organization's digital transformation. However, tax incentives can help to alleviate these constraints by increasing enterprise cash flow and lowering financing costs. As a result, tax breaks are critical in assisting enterprises with their digital transformation. As a result, when businesses rely heavily on external funding, financing becomes increasingly important. As a result, financing constraints increase, and the impact of tax incentives in driving the transition to digital intelligence becomes more apparent.

Therefore, this paper employs corporate external financing calculated as the ratio of net cash flow from investment minus net cash flow from operations total assets. Furthermore, values greater than the year-industry median indicate a high degree of external financing dependence (EF = 1), whereas values lower than the year-industry median indicate a low degree of external financing dependence (EF = 0). The results are displayed in the columns (1) and (2) of Table 7. The estimated coefficients are significantly positive at 1 %. However, the absolute value of the coefficients is higher in firms with a higher level of external financing dependence, which may suggest that implementing a tax reduction incentive policy has a greater impact on promoting digitization and intelligence in firms with stricter external financing dependence.

Table 7
Heterogeneous analysis results.

	EF = 1	EF = 0	control = 1	control = 0	dig = 1	dig = 0
	(1)	(2)	(3)	(4)	(5)	(6)
	digital	*digital*	*digital*	*digital*	*digital*	*digital*
tax	0.349***	0.193***	0.406***	0.016	0.278***	0.303***
	(0.064)	(0.060)	(0.075)	(0.055)	(0.058)	(0.064)
cons	1.294**	0.650***	1.231***	−0.574	1.245**	1.178**
	(0.569)	(0.054)	(0.061)	(0.468)	(0.505)	(0.569)
controls	YES	YES	YES	YES	YES	YES
FE	YES	YES	YES	YES	YES	YES
N	10,771	10,863	5416	16,220	11,566	10,070
R^2	0.256	0.280	0.299	0.174	0.195	0.262

Note: The robustness standard errors of the coefficients are in parentheses; ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

5.2.2. The mediation of industry nature

Regulated industries primarily include sectors critical to a country's well-being, national security, and the management of natural resources. These businesses heavily rely on government policies and are guided by specific policy goals. Regulated businesses have a stronger connection to the policy system than other industries, making achieving the desired outcomes from policy instruments easier. Enterprises in these industries have a rigid business model, limited awareness of intense market competition, and a need for greater motivation to engage in digital transformation. Policy tools, such as tax incentives, can mitigate the weak motivation of these enterprises to pursue digital transformation. Furthermore, these tools are critical for facilitating the digital transformation of regulated enterprises. Organizations in competitive industries prioritize digitization to improve production efficiency and maintain competitiveness. This motivation for change is intrinsic and does not depend on any financial incentives they may receive. This report posits that tax incentives significantly influence the digitalization of firms in regulated industries.

Therefore, this study divides the sample into businesses subject to regulation (control = 1) and industries (control = 0) operating in a competitive environment to perform regression analysis. Columns (3) and (4) of Table 7 show the results. The estimated coefficients are only significantly positive at 1 %, verifying the inference.

5.2.3. The mediation of urban digital infrastructure

This study divides the sample into developed (dig = 1) and underdeveloped (dig = 0) digital transitions based on urban infrastructure. The 50th percentile is used as the threshold to assess the moderating impact of urban infrastructure. Columns (5) and (6) of Table 7 report the results. The estimated coefficients are significantly positive at the 1 % level, but the value for firms with undeveloped digital infrastructure is greater than that of developed ones. The finding offers a practical guide for connecting the digital transformation of businesses in regions lacking digital development.

6. Conclusion

The emergence of digital intelligence has given organizations a significant opportunity to develop a new and distinct competitive advantage. Based on A-share listed firms from 2016 to 2022, this study adopts the DID method to explore the relationship between tax reduction incentives policy implementation and firms' digitization and intelligence. The finding reveals that implementing tax reduction incentives policy can effectively increase firms' digitization and intelligence. Furthermore, the positive effects are realized through resource complementarity and expected cash flow mechanisms. Tax reduction incentives have promoted businesses' digital transformation by encouraging human capital improvement and providing financial support through resource augmentation. Simultaneously, the tax reduction incentive policy stabilizes enterprises' anticipated cash flow, alleviates financing constraints, and provides fundamental assurance for the implementation and advancement of digital technology, facilitating the transition to digital intelligence for enterprises. Moreover, firms' external financing dependence, industry nature, and the extent of urban digital infrastructure have heterogeneous mediating effects. The study's findings have important practical implications for the corporate digital and intelligent transition.

Therefore, the following policy implications are proposed. First, intelligent transformation is a structured investment initiative involving human-machine collaboration and hardware and software integration. Implementing VAT reduction reform and income tax incentives has boosted fixed asset and software investment from several perspectives. Large-scale organizations and small enterprises have different investment patterns for intelligent transformation. As enterprises undergo intelligent upgrades, investments in software and services will increase. The current tax reduction strategy prioritizes hardware investment, and it should effectively encourage strategic investment in intelligent transformation. Thus, it is more advantageous to implement specific tax incentives for projects involving intelligent investment. The specialized tax reduction strategy has a more defined incentive goal, simplifying the tax department's operation. It also includes regularly evaluating the policy's impact, allowing for timely policy adjustments. Enterprises have the advantage of choosing systematic decision-making alternatives for intelligent project planning, which increases flexibility. Intelligent investment projects are eligible for various tax breaks, including VAT and income tax rebates. The primary goal of the VAT policy is to implement the tax retention and refund policy. Meanwhile, the income tax policy can be tailored to the unique characteristics of various investments. It can effectively integrate various preferential methods, such as investment tax credits, additional expense deductions, and tax deductions. Furthermore, the tax department should set parameters for targeted investment in implementing intelligent systems. This investment should include the strategic allocation of resources to intelligent hardware, software, and network infrastructure, as well as staff training and recruitment of skilled personnel.

Second, improving the integration and coordination between fiscal and tax reduction policies and optimizing the impact of the policy combination are crucial. Fiscal incentives and subsidies for pilot initiatives focused on intelligence and digitization, which have a limited field of application, are highly advantageous and provide an additional benefit to firms. The tax reduction proposal requires more specificity, and it will benefit a wider range of people. Furthermore, tax reduction policies provide significant incentives for investment in intelligent transformation; however, no specific tax reduction policies focus on the contribution of intelligent production. This disparity may result in excessive investment. An optimal policy configuration would include developing tax reduction policies specifically for intelligent transformation. Under this arrangement, firms facing significant challenges would receive financial incentives for various projects. Fiscal incentives are available for projects that significantly affect intelligent

transformation and improve the manufacturing industry's international competitiveness. These projects include advances in cutting-edge technology, enterprise export substitution, innovative outputs that stimulate demand, and transformation initiatives that result in greater efficiency gains. Moreover, fiscal subsidies and incentives complement tax-cutting initiatives, allowing the combined effect of the policy mix to be more effectively realized.

Although this study provides valuable insights, some limitations necessitate further research. The sample solely comprises listed Chinese enterprises, which may introduce bias in the selection process. Therefore, future studies should use sophisticated econometric methods to address these challenges and consider expanding the research domain to include other developing economies.

Funding

grants from the National Social Science Foundation of China (22&ZD089).

Data availability

The authors do not have permission to share data.

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